

DermaVision System **(Multimodal Skin Disease Classification)**

Meta Learning Late Fusion & CNN-Based Clinical Diagnosis using TF

Presentation Roadmap

A comprehensive overview of our research, methodology, and results.

AI in Modern Healthcare

Transforming Diagnosis

The integration of machine learning into clinical workflows is revolutionizing early detection. In dermatology, automated analysis provides a critical second opinion, reducing diagnostic variability and improving patient outcomes in low-resource settings.

The Diagnostic Challenge

Core Bottlenecks

Medical diagnosis faces hurdles in data consistency. Small medical datasets often lead to overfitting, while heterogeneous data types (images and text) are difficult to process simultaneously in standard models.

Resource Constraints

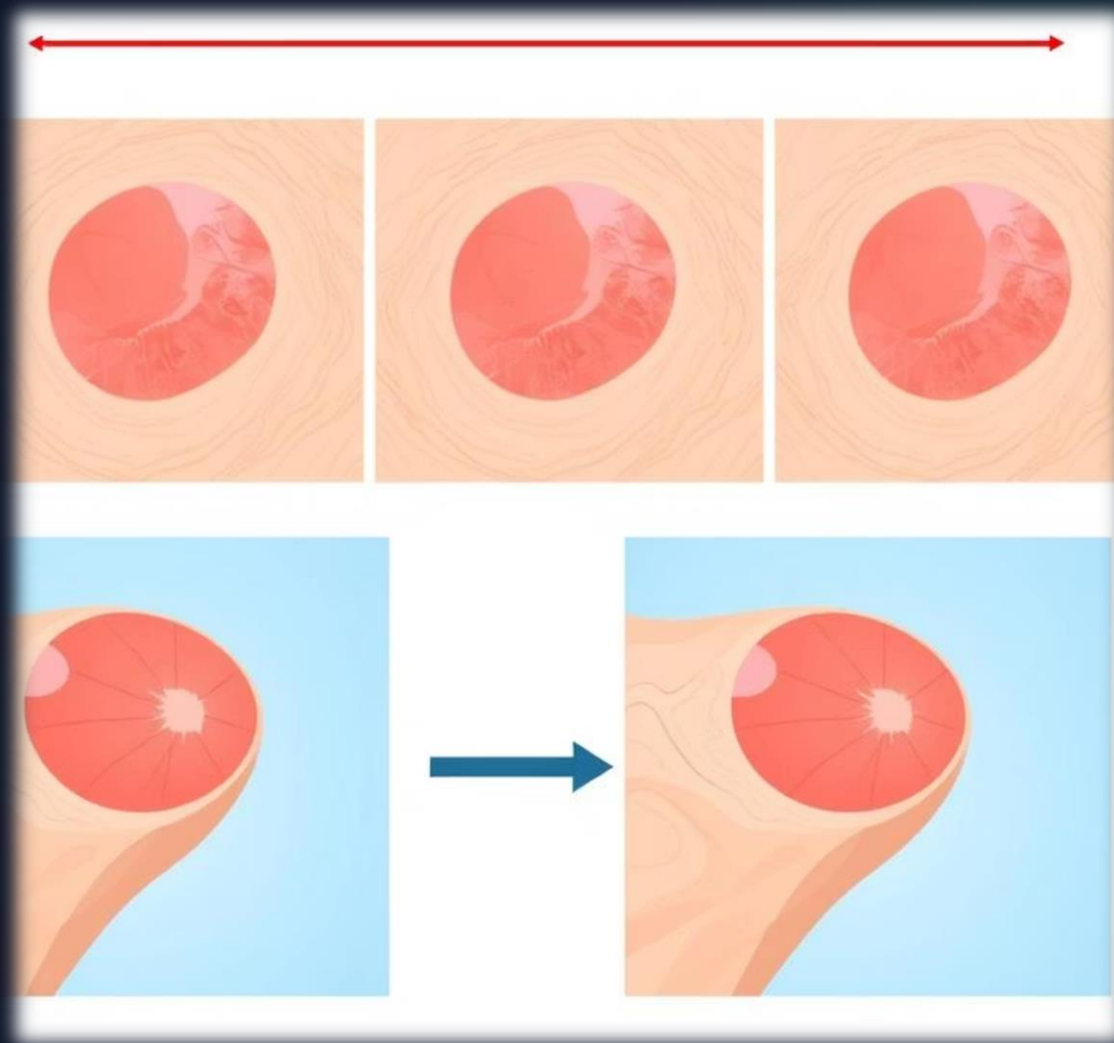
Traditional biopsy-based diagnosis is time-consuming and expensive. There is an urgent need for robust, multi-modal screening tools that can leverage both visual and clinical evidence.

The Multi-modal Solution

Strategic Feature Fusion

Our approach innovates by merging visual features extracted via CNN with structured patient metadata. By creating a unified representation, the system captures correlations between image patterns and clinical history.

- 🔧 CNN Image Analysis
- 🔧 ANN Feature Analysis
- 🔧 Ensemble Late Fusion



The PAD-UFES-20 Dataset

Exploring the core data foundation for skin lesion classification.

Clinical Metadata Depth

22 Patient Features

Beyond images, the dataset includes rich clinical context. These features are critical for distinguishing between lesions that appear visually similar but have different clinical risk profiles.

Diseases: BCC, ACK, NEV, SEK, SCC, MEL.

Key Metrics: Patient age, gender, lesion location, skin type, and lesion diameter.

Project Execution Roadmap



Preprocessing & KNN Imputation

Missing Value Handling

Medical records often contain 'UNK' or 'NaN' values. We utilized **KNN Imputer** to estimate these values based on the similarity of nearest neighbors, preserving dataset size.

Normalization & Encoding

Numerical features (Age, Size) were Standard Scaled. Categorical features (Sex, Race, Region) were One-Hot Encoded to ensure model mathematical compatibility.

Strategic Augmentation

Expanding the visual horizon to prevent model overfitting.

The Small Dataset Challenge

9,192

Total Images After Augmentation

Artificial Diversity

Starting with only 2,298 images is insufficient for deep CNN training. Overfitting occurs when the model memorizes training noise rather than general patterns.

Impact: Augmentation quadrupled our dataset, providing the necessary variance for clinical robustness.

Phase 1: Geometric Flips

Horizontal & Vertical Inversion

By flipping images across axes, we ensure the model does not associate specific diseases with a specific corner or orientation of the frame.

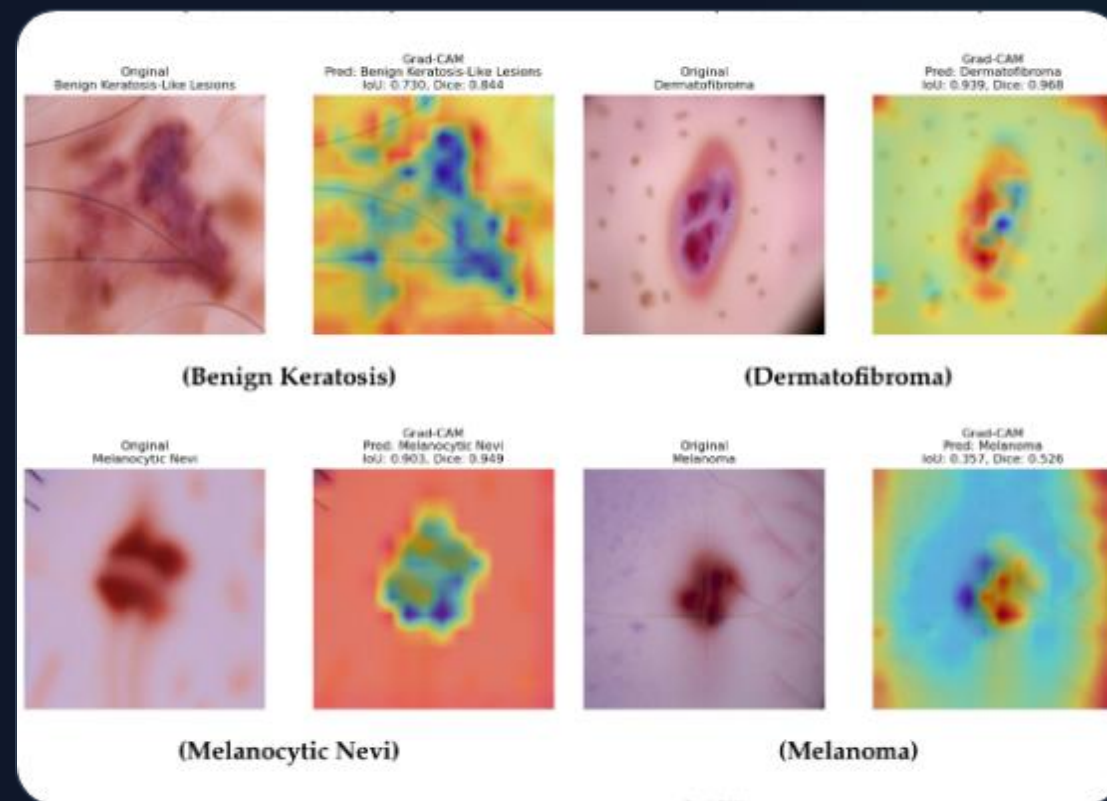
Result: Created 2,298 additional perspectives, forcing the model to focus on invariant lesion structures, we have 4596.

Phase 2: Noise & Color Shift

Gaussian Noise & Color Shift

Real-world clinical images vary in lighting and camera sensor quality. Injecting random noise and adjusting saturation prepares the model for these environmental inconsistencies.

Benefit: Improves performance on unseen "messy" data from different smartphone models, we have 6894.



Phase 3: Scale & Rotation

Zoom & Random Rotation

Lesions appear at different distances and angles. Applying a 0.8x to 1.0x zoom and 15-degree rotations simulates varying dermatoscopy techniques.

Final Dataset: 9,192 unique observations ready for the CNN training pipeline.

Architecture & Training

Building the visual intelligence engine.

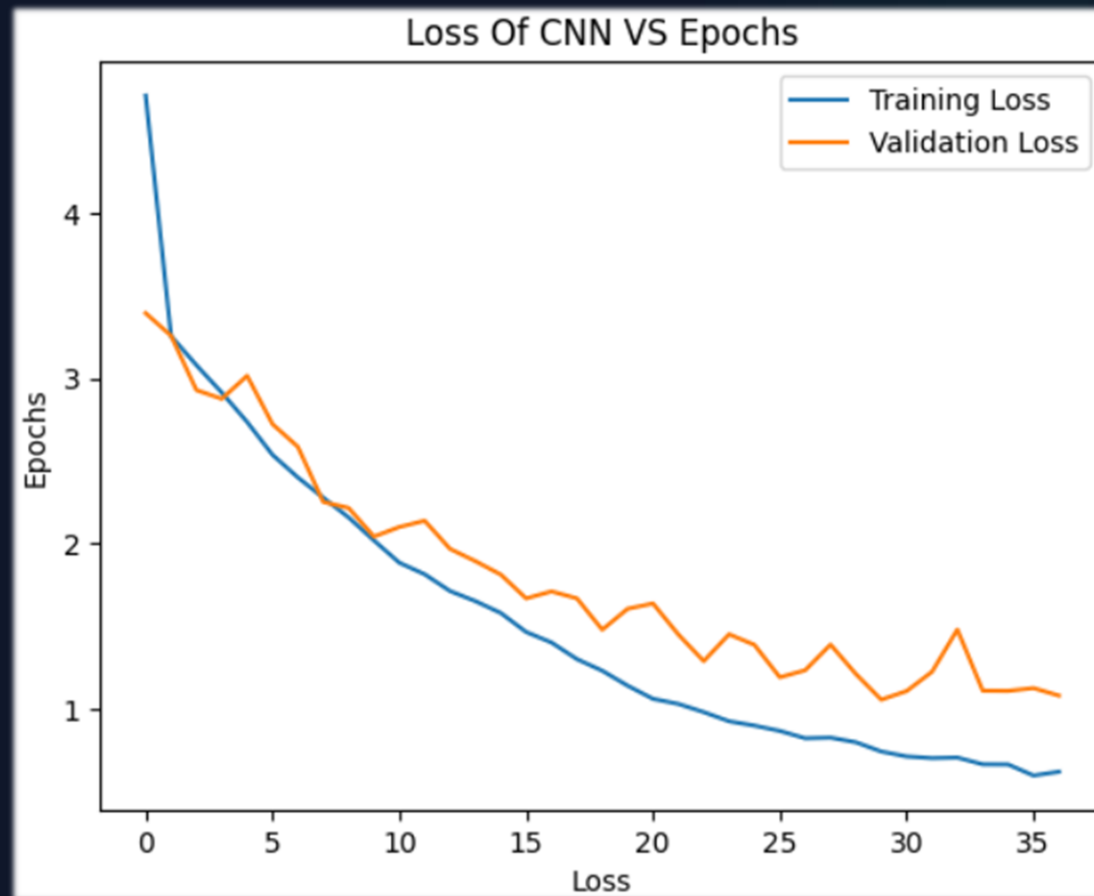
CNN Engine & Baseline

Initial Training (56%)

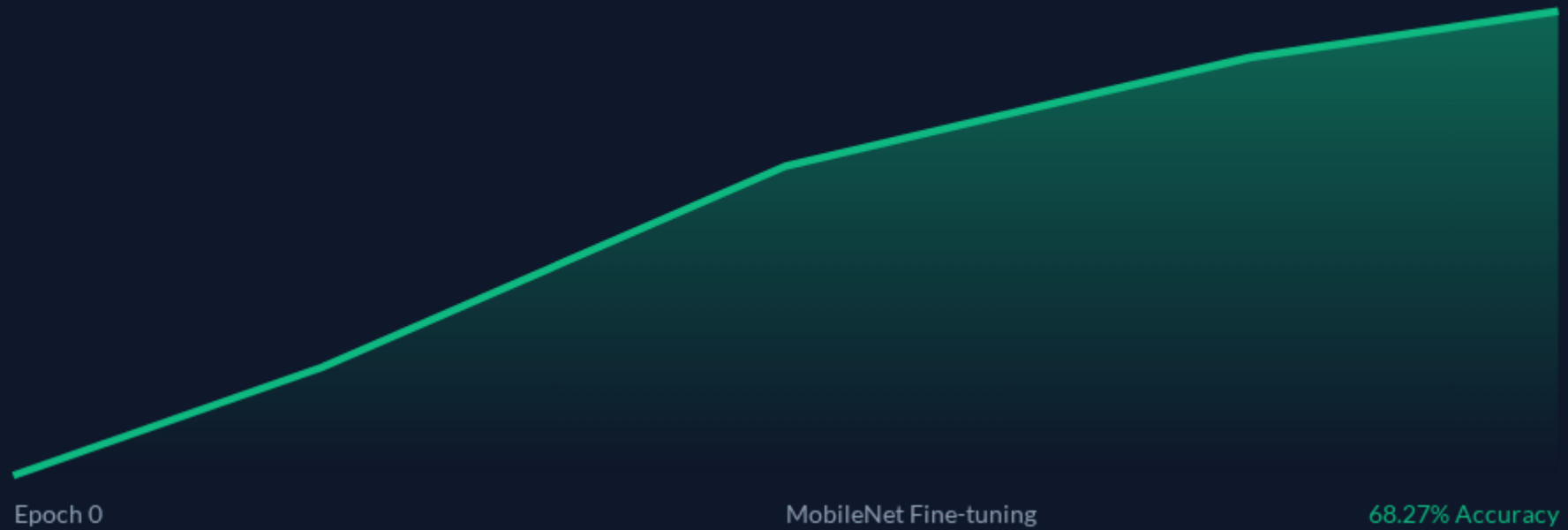
Our custom CNN, trained from scratch on 9,192 images, established a baseline accuracy of 56%. This confirmed visual features alone are insufficient for high-precision skin cancer diagnosis.

The Underfitting Issue

Scratch models struggle to learn complex edge and texture features from limited medical datasets. This necessitated the transition to **Transfer Learning**.



The Accuracy Leap



Utilizing pre-trained MobileNet features accelerated convergence and boosted visual accuracy.

ANN Engine

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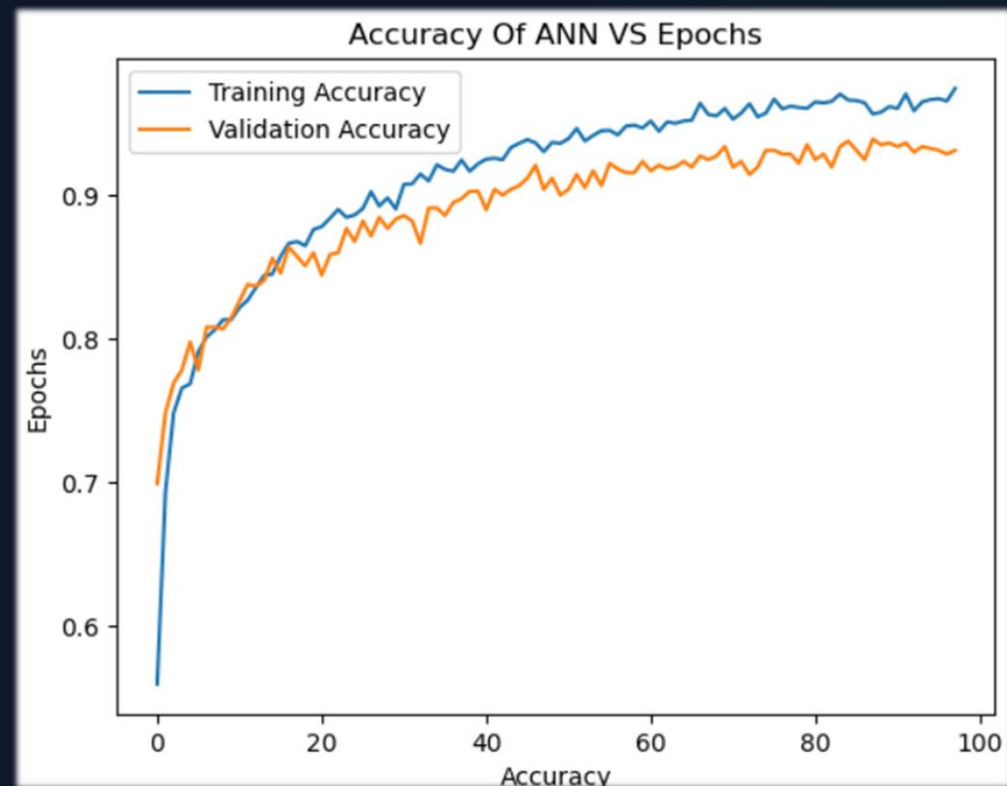
ANN Engine

ANN For Features

We train ANN for features of the patient, then analysis the data of it.

The number of samples was 9192.

We get final accuracy 93.35%.

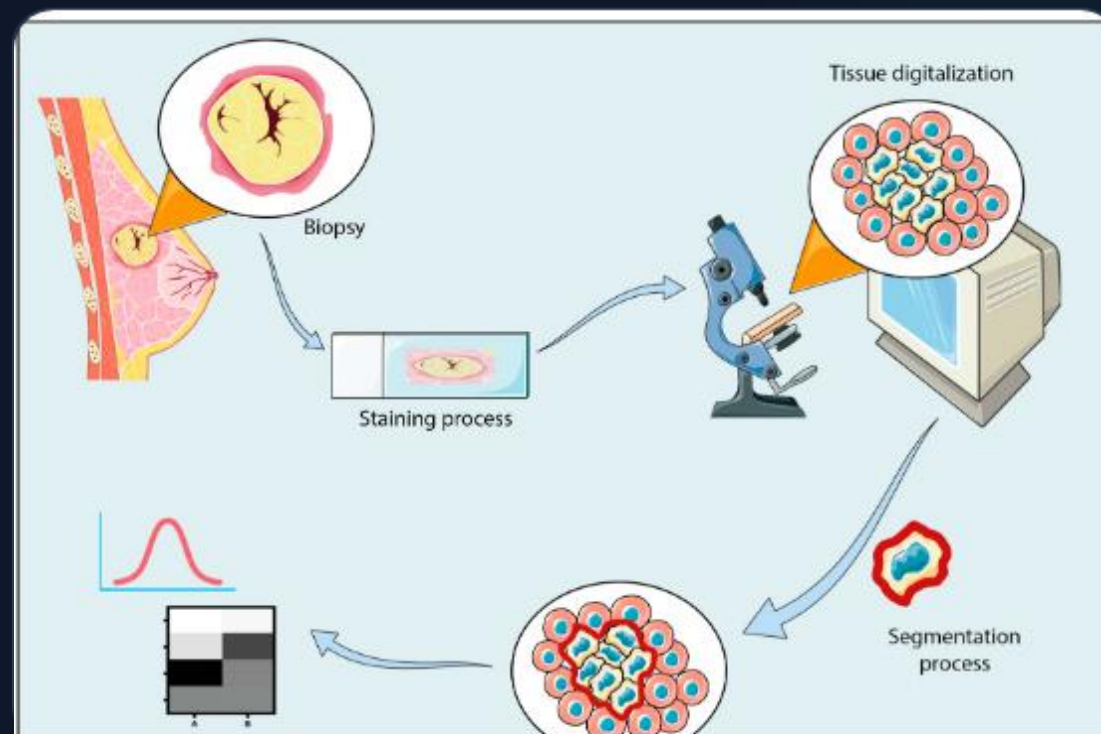


Visual Probability Mapping

From Image to Vector

Instead of hard labels, we extract the final softmax layer's **probability vector**. This captures the model's confidence across all 6 classes.

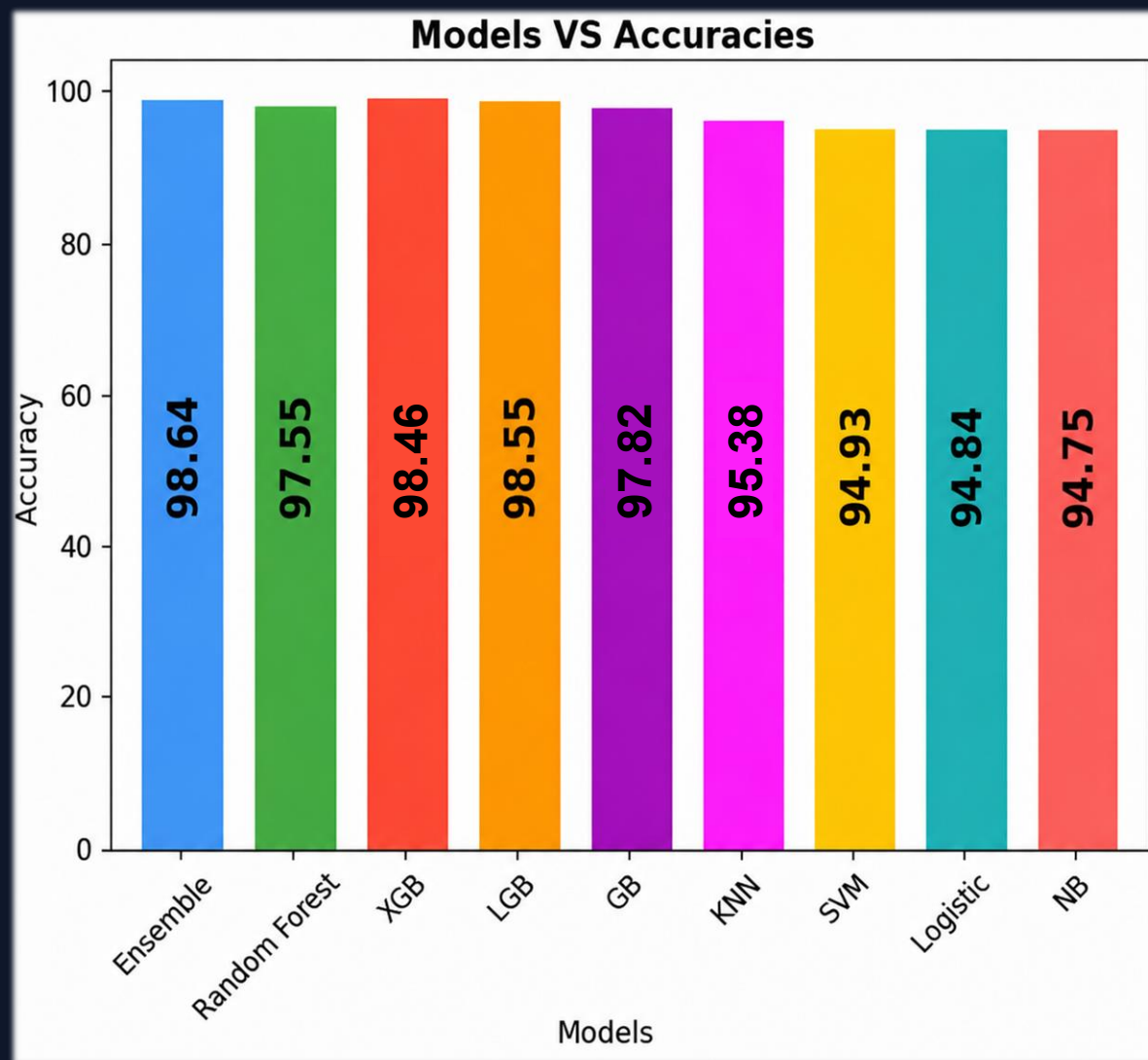
This 6-dimensional vector becomes the 'visual signature' that is passed to the fusion layer.



Advanced Data Fusion

Merging visual confidence with clinical evidence.

Classifier Benchmarking



Ensemble Performance

98.64%

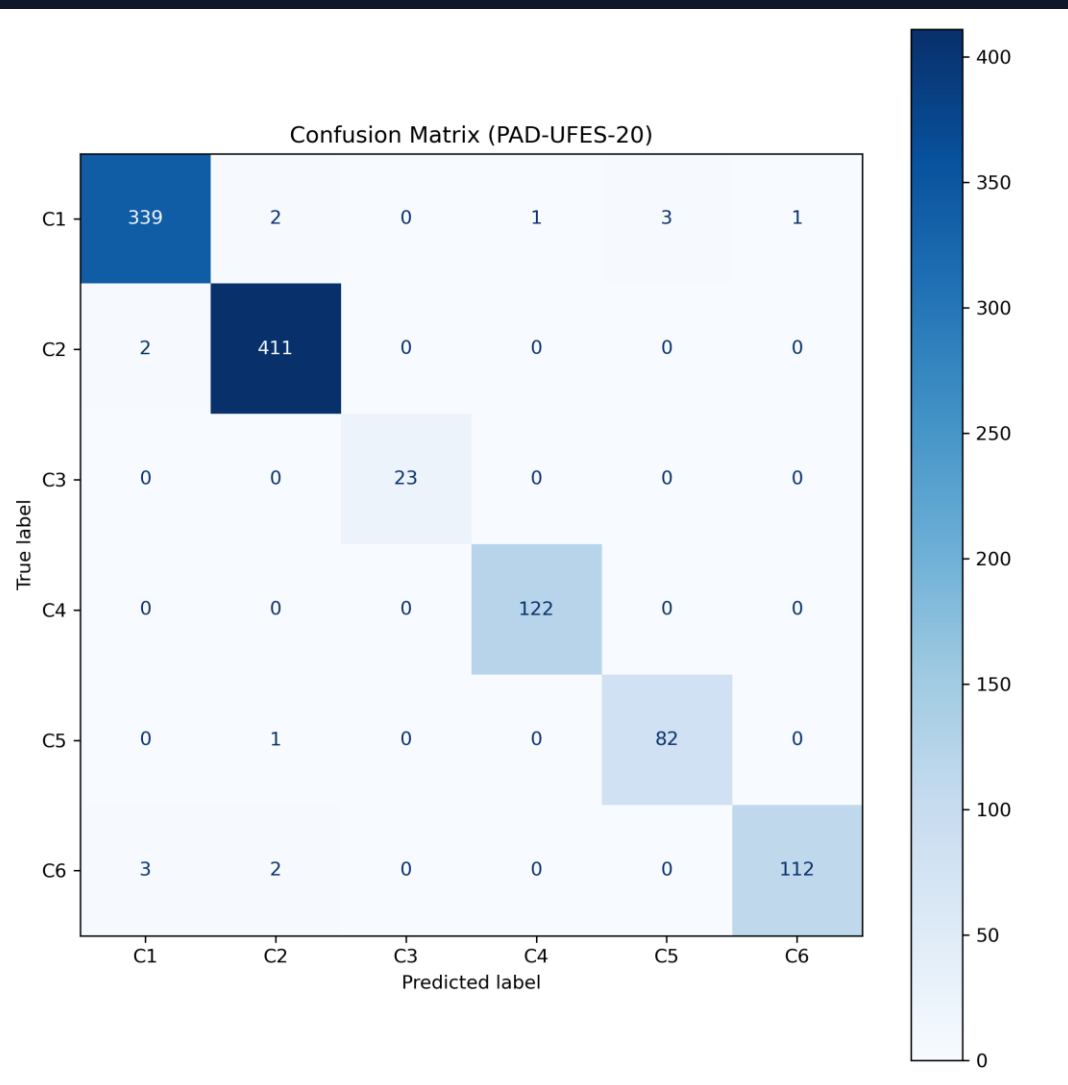
Final System Accuracy

Power in Numbers

By combining the top 3 performing models (XGB, LGB, GB) into a weighted ensemble, we maximized robustness.

The ensemble approach ensures that the system is not vulnerable to the specific biases of a single classifier.

Confusion Matrix Insight



Confusion Matrix

The high diagonal density confirms minimal misclassification across all classes. Notably, high-risk cancers (Melanoma, SCC) are identified with near-perfect sensitivity, ensuring no dangerous lesions are missed.

Flask Web App: Practical Application

Project Link

🔬 AI-POWERED DERMATOLOGY

Skin Disease Prediction System

Upload a dermoscopy image and complete the patient questionnaire to receive an AI-assisted diagnosis.

STEP 0 — PATIENT & IMAGES

Enter patient details and upload images to begin

👤 PATIENT FULL NAME *

IMAGES

📎 LESION IMAGE *



Click to browse or drag & drop

PNG · JPG · JPEG — max 16 MB

👤 PATIENT PHOTO (OPTIONAL)



Click to browse

Optional · PNG · JPG

Next → Start Questionnaire

| Project Files

GitHub Repo



[GitHub Repo](#)

Dataset Link



[Dataset Link](#)

Questions & Answers

Thank you for your attention. Feel free to ask about our methodology, the dataset, or deployment strategies.